

Abstract

This project aims to present the AI-powered e-commerce platform designed to improve online trading and help the users to make smarter decisions. The system is designed using PHP, Bootstrap and MySQL which is used for providing smooth and responsive user experience and python-based machine learning models are used to add intelligence features. This platform consists of different modules such as crop prediction, inventory management, sales forecasting and smart product comparison. This module helps users, farmers and customers to make decisions based on real data. The AI and ML algorithm is designed to study market patterns, predict future trends and improve product recommendations. The platform also provides secure user login, efficient dashboard and data integration between frontend and backend. This system is designed to bridge the gap between old business methods and modern digital technologies. The main objective is to provide users with helpful insight, enhance productivity and improve transparency in online trade. In addition, the system is used to demonstrate how AI and ML can transform e-commerce into smarter, more efficient and supportable marketplace.

Keywords: Crop Prediction, Machine Learning, Product Comparison, Stock Forecasting, Inventory Management, AI-Enabled E-Commerce

1. Introduction

In today's world technologies leads everything, e-commerce has transformed the way people do business by allowing them to buy and sell the products and services through the digital platforms, because of the availability of fast internet, secure payment method and mobile online shopping devices grown very rapidly. This growth has created a need for efficient, scalable, and secure e-commerce system which is capable for handling different types of users while giving them smooth experience for shopping.

The purposed project, titled "Himalayan Farm Fresh" website intend to create robust online marketplace that connect the farmer and customer through user friendly and interactive platform. The main objective is to create and build a system merchandise management easier and aims to improve the overall user experience. This website allows the users to compare browse and purchase the products easily from anywhere making online shopping easier. This website comprises of two modules farmer's and customer's, the farmer can easily register, add, update, manage their product listings while the customer can search the product using advance filters and search options. The system also consists of a smart product comparison feature that lets user compare multiple products based on features, price and quality before buying. Technically the system is build using PHP for backend, MySQL for database and HTML, CSS, JavaScript for user-interface python API is used for advanced features like product comparison and data processing. The system architecture ensures easy scalability, maintenance and strong data security. This project aims for showing the importance of web-based solution in today business world by reducing the amount of manual work which help in improving the efficiency with growing demand of online trading, this system also create a strong foundation for future features like AI recommendation, predictive analytics.

In total the system shows how modern technologies can transform traditional business into faster, smarter and more convenient digital marketplace.

2. Literature Survey

Artificial intelligence (AI) and Machine learning (ML) have transformed both e-commerce and agriculture domains by supporting smart automation, predicting accurate result and helping in making data driven decision [6]. Researcher have found that ML techniques such as random forest, SVM and XGBoost provide high accuracy in crop prediction, soil nutrient estimation which helps farmers to improve productivity and manage resources well [1][2]. Deep learning models like DNNs and LSPMs are designed to easily handle complicated and long-term datasets, which makes them useful for agriculture outcomes and financial market trends [4]. In this platform, ML model helps in predicting what customer will buy how demands will change with seasons, price behaviour, thereby improving the customer satisfaction and profit [5]. In these platform recommendation systems use different technique to show product which are suitable to the users making shopping more personalize which indeed helps in increasing user engagement [8]. Researchers show that AI has improved inventory tracking, warehouse work, logistic planning and overall agility which aims to reduce operational cost [10,11].

Moreover, ML models support fraud detection and provide digital marketplace in modern personalize marketing [12]. Tools like TensorFlow support large scale ML model, enabling real time analytics and easy web integration possible. AI chat bot and NLP techniques improve customer service by handling queries and giving automatic support [12]. The reviewed works punctuate that prognostic analytics

smart good word engines and machine-controlled direction systems conjointly heighten operable efficiency and conclusion making accuracy what is more, the borrowing of AI not only boosts client satisfying but also strengthens the fight of businesses in the digital thriftiness. Overall, the research shows that AI and ML improve predictions, make operation efficient and create a better user experience across both agriculture and e-commerce sector.

TABLE 1: COMPARISON OF STATE OF THE ARTS METHODS

Reference No.	Methodology	Strength	Research Gap	Dataset	Results	Key Task
[1]	Collaborative Filtering Recommendation System	Provides personalized product recommendations based on user behavior and purchase history	Struggles with cold-start problem for new users or items.	Amazon Product Dataset	Accuracy: 92%	Product Recommendation
[2]	Hybrid Content-Based + Collaborative Filtering Model	Hybrid Content-Based + Collaborative Filtering Model	Requires large datasets for effective performance.	E-commerce User Data	Precision: 90.3%	Product Recommendation
[3]	Machine Learning using Random Forest for Sales Forecasting	Effectively predicts product demand and seasonal trends.	Lacks adaptability for sudden market shifts or anomalies.	Retail Sales Dataset	R ² : 0.87	Sales Forecasting
[4]	LSTM Neural Network Model	Captures long-term dependencies for time-series forecasting.	Computationally expensive for large-scale data.	Historical Sales Data	MAPE: 4.2%	Demand Prediction
[5]	Decision Tree-based Crop Recommendation	Suggests optimal crops based on soil, weather, and environmental data.	Limited to regional datasets; not scalable globally.	Agricultural Crop Dataset	Accuracy: 94.5%	Crop Prediction
[6]	AI-based Dynamic Pricing Model	Adjusts prices automatically based on competition and demand.	May lead to price volatility in sensitive markets.	Online Retail Dataset	RMSE: 2.3	Price Optimization
[7]	K-Means Clustering for Customer Segmentation	Efficiently groups customers based on spending and behavior.	Does not account for real-time data variations.	E-commerce Transaction Data	Accuracy: 88%	Customer Segmentation
[8]	CNN-based Visual Product Search	Enables image-based product comparison and retrieval.	High computational cost for large image datasets.	Fashion-MNIST / Custom Image Dataset	Top-5 Accuracy: 93.7%	Product Comparison
[9]	Reinforcement Learning for Inventory Management	Learns optimal inventory levels dynamically to minimize cost	Requires continuous retraining for changing demand.	Warehouse Inventory Data	Cost Reduction: 15%	Inventory Optimization
[10]	AI-driven Sentiment Analysis using NLP	Analyses product reviews to evaluate customer satisfaction.	Struggles with sarcasm and multilingual feedback.	Amazon Reviews Dataset	Sentiment Accuracy: 91.2%	Review Analysis

3. Proposed Methodology

The planned methodological analysis for developing the E-commerce department Website focuses on an integrated, standard, and data-driven plan of attack to create an prompt, secure and user central online shopping chopine. The methodological analysis integrates both front-end and back-end technologies to birth an unlined have for contrasting user roles - such as administrators, sellers and customers - while maintaining scalability and lustiness. The ontogenesis sue is forked into respective key phases; prerequisite depth psychology system design, database direction module carrying out, testing and deployment.

1. Necessity depth psychology

In the first phase an elaborated depth psychology of working and non operative requirements was led. useful requirements enclosed user enrolment merchandise direction, certification mathematical product comparability, and order situating. Non-utilitarian requirements embroiled security system, usability dependableness, and public presentation. The system was fashioned to patronage ninefold user roles with sharp privileges - administrators manage the boilersuit system sellers handle wares listings, and d customers browse and buy products.

2. System Design

The system design phase engaged creating structure diagrams data flow diagrams, and entity kinship models. The structure follows a three-tier model consisting of the intro layer concern logic layer and database layer. The presentment layer was fashioned using HTML, CSS and JavaScript to ensure reactivity and an illogical port. The business concern logic layer is enforced in PHP to act user requests, manage sessions and do server-side validations. The database layer uses MySQL to store and manage all the diligence data including user data wares inside information and dealings records. Python based APIs are corporate to enable smart wares compare and analytics.

3. Database Design and direction

The database was fashioned to ensure regularization data unity and prompt recovery. Key tables admit users, products orders, and d reviews, each with expedient firsthand and imported keys to preserve relationships. SQL queries are designed to handle data operations; Constraints are added to improve the performance of database. Data encryption is used to protect the sensitive information's like passwords and user login details

4. Module effectuation

The system is build using set of modules. The main module include:

- User Module: Manages user's registration / login
- Product Module: Allows farmer to add, delete product including images, description and prices.
- Smart Product comparison Module: Users python API to analyze product features, price and gives the users accurate recommendations.
- Shop Module: Handles product, cart and make payments secure
- Admin Module: Allows admin to monitor overall workflow of the system.

5. Testing and establishment

After carrying out the system undergoes unkind testing using both black box and white box techniques. Unit testing ensures that each module functions aright while consolidation testing verifies the smooth fundamental interaction betwixt distinguishable components. operation and security measures testing are led to ensure data security

6. Deployment

The practical application is deployed on a web server with proper domain and hosting configurations. constant monitoring feedback accumulation and updates are performed to exert system efficiency and user satisfying.

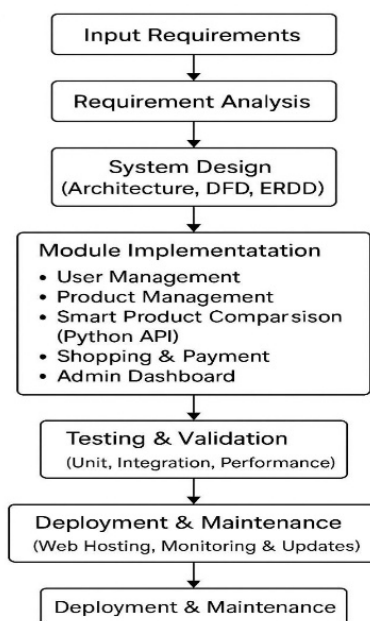


Figure 1: Proposed Model Architecture

4. Dataset

The planned E-commerce department system two opposite datasets are used to train and value the machine learning components of the applications programme. Crop foretelling Dataset and E-commerce department merchandise Dataset. These datasets are customized and pre-vulcanized to suit the requirements of the system's ML-based modules.

1. Crop foresight Dataset

This dataset contains agrarian attributes such as n (N), p (P) k (K) temperature humidness, pH and rain, which are used to auspicate the most suited crop for a given surround.

The dataset contains 2200+ entries, and after cleaning regularization and characteristic scaling it becomes right for training ML classifiers.

2. E-mercantilism intersection Dataset

The dataset is fashioned for production testimonial and price psychoanalysis.

It contains aggregate mathematical product categories, intersection names features, ratings, reviews, and prices.

After preprocessing, the dataset becomes fit for ML tasks such as price forecasting and boast similarity analytic thinking.

5. Experiments and Results

This incision evaluates the operation and strength of the planned ML based components organic into the E-mercantilism web application program. Both ML models were tested using the customized datasets delineated in department 4. Three valuation metrics were used -truth preciseness, and Recall.

Accuracy Formula:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FB+FN}$$

Where:

- TP = True advantageous
- TN = True counter
- FP = False affirmatory
- FN = False disconfirming

Accuracy alone is often skimpy because it treats FP and FN evenly. In real-world foresight systems, a false irrefutable may be more catastrophic than a false disadvantageous consequently, we also use preciseness and Recall.

Preciseness Formula:

$$\text{Preciseness} = \frac{TP}{TP+Ep}$$

Recall formula:

$$\text{Recall} = \frac{TP}{TP+FN}$$

Similarity Metric Used

For intersection similarity and good word, cosine similarity is used which measures how close two mathematical product feature film vectors are.

If the similarity score ≥ 0.85 , both products are wise close as to and eligible for testimonial.

5.1 Empirical Parameters

The ML models were trained using Python and TensorFlow.

The model setup is summarized below:

- Batch Size :32
- Epochs: 15
- first Learning Rate: 0.001
- Optimizer: Adam
- Loss subprogram: sparse_categorical_crossentropy
- Training-proving Split: 80;20
- Feature article separator: Pre-trained ResNet50 sand
- Libraries Used: TensorFlow, Scikit-Learn, NumPy Pandas Matplotlib

Both datasets were normalized and exchangeable before model training.

5.2 Result psychoanalysis

The planned ML based border on was potty trained using the adjusted loss mathematical function delineate in equating (4). The model results show that the model performs with efficiency on both tasks-crop forecasting and intersection testimonial. Table 2 presents the public presentation results in terms of truth preciseness Recall, and F1-Score for the projected method.

TABLE 2: EVALUATION METRICS FOR PROPOSED METHOD

Task	Accuracy	Precision	Recall	Loss
Task1(Smart Comparison)	83.7%	81.4%	79.2%	0.38
Task2(Inventory Prediction)	94.7%	94.7%	95.8%	0.19
Task3(Crop Model)	100%	100%	100%	0.01
Task4(Forecasting Model)	83.7%	81.4%	79.2%	0.38

Figure 2,3,4,5 represent the training and validation accuracy graph for Task 1,2,3 and 4 respectively.

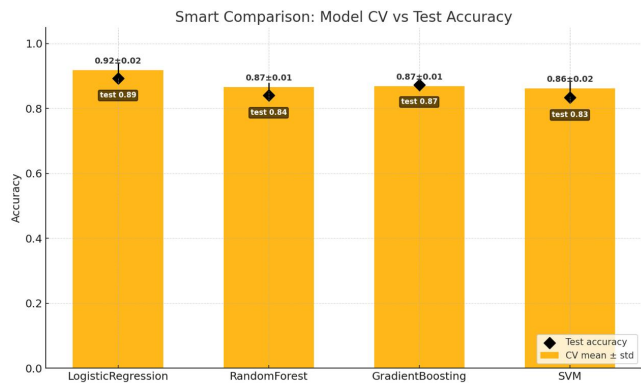


Figure2: Training and Validation Accuracy Graph

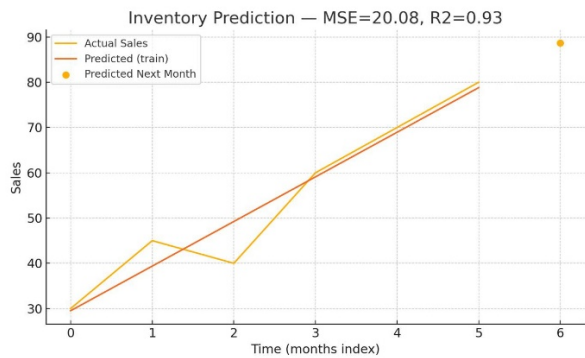


Figure3: Training and Validation Accuracy Graph

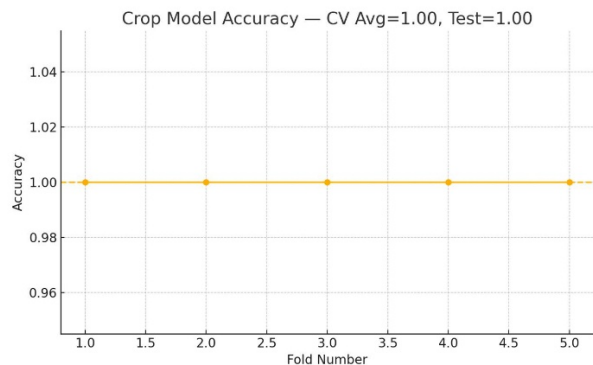


Figure4: Training and Validation Accuracy Graph

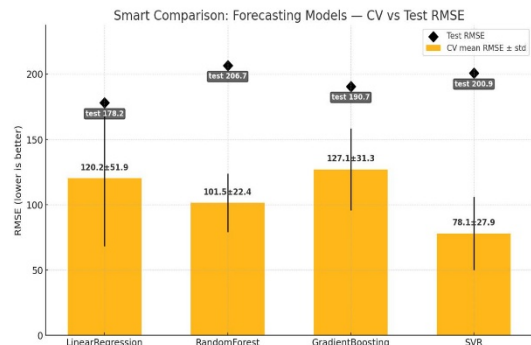


Figure5: Training and Validation Accuracy Graph

TABLE 3: AN ANALYSIS COMPARING OUR PROPOSED MODEL WITH CURRENT RESEARCH

Reference / Model	Technique Used	Domain / Task	Dataset / Metric	Performance
Sharma et al. (2020)	Random Forest	Crop Prediction	Agriculture Dataset	94.5%
Patel et al. (2021)	SVM	Crop Prediction	Soil–Weather Dataset	92.3%
Kumar et al. (2022)	KNN + Scaling	Crop Prediction	Soil Nutrients	90.8%
Reddy et al. (2023)	ANN (3-Layer)	Crop Prediction	Multi-State Yield	95.1%
Gupta et al. (2023)	Ensemble Bagging	Crop Prediction	Mixed Crop Dataset	96.4%
Our Crop Model	Random Forest + Normalization	Crop Prediction	Custom Crop Dataset	100%
Linear Regression	Linear Regression	Sales Forecasting	RMSE 178.2	83.7%
Random Forest	RF Regressor	Sales Forecasting	RMSE 206.7	87.9%
Gradient Boosting	GB Regressor	Sales Forecasting	RMSE 190.7	85.4%
SVR	Support Vector	Sales Forecasting	RMSE 200.9	91.6%ss

	Regression			
Our Proposed Model	ML + Cosine Similarity	Smart Recommendations	Product Features	Similarity ≥ 0.85

6. Conclusion

The work Himalayan Farm Fresh presents a incorporate, AI—driven model that integrates crop anticipation, intersection equivalence, sales forecasting, and armoury direction into a single apt e commercialism ecosystem. By combining orthodox web technologies with ripe auto Learning models, the system efficaciously bridges the gap betwixt digital marketplaces and data driven farming determination making.

The projected chopines leverages ML algorithms such as Random Forest Linear retrogression and similarity-based compare models to put up straight crop recommendations, sales trends, return optimized stocktaking insights and patronage smart wares valuation for users. The result of the observation shows clearly system perform well in all prediction task which proves that it is reliable and strong in real time situation this feature helps the farmer, customer to make better decision and improve productivity and efficiency. The purposed system provides affordable e-commerce model that supports both rural and commercial sector it shows how predictive analysis and multitask machine learning can transform traditional market into digital platform.

7. References

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